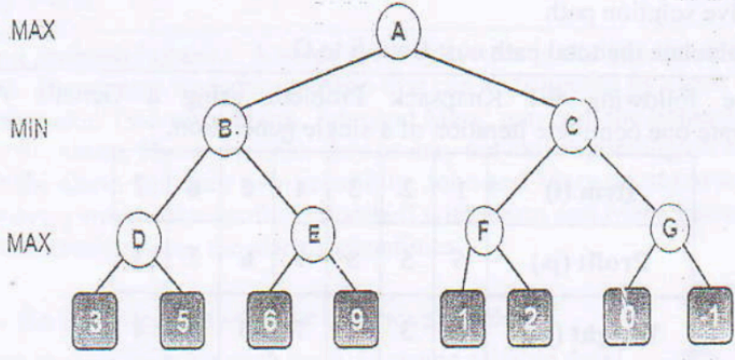
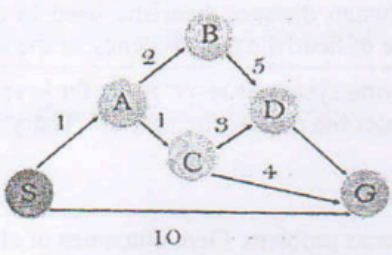


Jan 25 - Apr 25

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	Discuss any five limitations of AI.	5
ii)	Define Heuristic. What is the Manhattan distance heuristic used in an 8-puzzle problem? Discuss the effect of choice of heuristic on efficiency of the solution.	1+2+2
iii)	Consider a scenario - A garden watering system is responsible for keeping the plants hydrated. The system must water the plants when the soil is dry and stop when it's wet. Give PEAS analysis for this problem.	5
iv)	Give problem formulation for N-Queens problem. Give outcomes of all steps.	5
v)	In the context of adversarial search: Define Max Define Min Define alpha value Apply MAXMIN algorithm to the following instance and compute the value at root node A. 	1+1+1+2
vi)	What is the key difference between an AI program and an expert system? Comment on significance of this differentiating component.	5

Que. No.	Question	Max. Marks																								
Q2 A	Solve the following	10																								
i)	State and justify in one line properties of task environment for the garden watering system mentioned above in Q. 1. iii.	5																								
ii)	Discuss the problems in the hill climbing process, state the solutions over the same.	5																								
OR																										
Q2 A	Suggest and justify suitable agent architecture to design the garden watering system mentioned in Q. 1 iii. Draw appropriate standard diagram and give sample contents for all blocks in the diagram.	10																								
Q 2 B	Solve any ONE	10																								
i)	Consider the following graph where: each edge has an associated path cost and heuristic value $h(n)$ is provided in the table. Start node is S and Goal node is G. <table><thead><tr><th>State</th><th>$h(n)$</th></tr></thead><tbody><tr><td>S</td><td>5</td></tr><tr><td>A</td><td>3</td></tr><tr><td>B</td><td>4</td></tr><tr><td>C</td><td>2</td></tr><tr><td>D</td><td>6</td></tr><tr><td>G</td><td>0</td></tr></tbody></table> Apply the A* algorithm starting from node S to reach node G. - Draw the State Space Tree for the given graph - Show calculation of heuristic value at every node along the path being explored. - Give solution path - Calculate the total path cost from S to G.	State	$h(n)$	S	5	A	3	B	4	C	2	D	6	G	0	10										
State	$h(n)$																									
S	5																									
A	3																									
B	4																									
C	2																									
D	6																									
G	0																									
ii)	Solve the following 0/1 Knapsack Problem using a Genetic Algorithm. Demonstrate one complete iteration of a single generation. <table><thead><tr><th>Item (i)</th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th><th>6</th><th>7</th></tr></thead><tbody><tr><td>Profit (p_i)</td><td>6</td><td>5</td><td>8</td><td>9</td><td>6</td><td>7</td><td>3</td></tr><tr><td>Weight (w_i)</td><td>2</td><td>3</td><td>6</td><td>7</td><td>5</td><td>9</td><td>4</td></tr></tbody></table> Knapsack Capacity (W) = 9 Clearly state: Fitness function, crossover points, mutation, etc. i.e. details of all steps.	Item (i)	1	2	3	4	5	6	7	Profit (p_i)	6	5	8	9	6	7	3	Weight (w_i)	2	3	6	7	5	9	4	10
Item (i)	1	2	3	4	5	6	7																			
Profit (p_i)	6	5	8	9	6	7	3																			
Weight (w_i)	2	3	6	7	5	9	4																			

Que. No.	Question	Max. Marks
Q3	Solve any TWO	20
i)	Consider following set of sentences: 1. Steve only likes easy courses. 2. Science courses are hard. 3. All the courses in the basket weaving department are easy. 4. BJ301 is a basket weaving course. Convert the following sentences into <u>First-Order Logic</u> (FOL). Using <u>Backward Chaining</u>, prove the statement "Steve likes BJ301." State all inference rules used clearly.	4 + 6
ii)	Convert the following sentences into FOL and then into CNF. [<u>Step by step conversion from FOL to CNF is not expected</u>] 1. Everyone has birthdays on some day. 2. Good dogs are always with their masters. 3. A barber shaves all men in the city who do not shave themselves. 4. Fred is a trained dog. 5. Every boy or girl is a child.	10
iii)	Consider an agent that delivers parcels from a warehouse to a customer within a specified time window / chosen time slot. a. Identify and explain the key uncertainties the agent may encounter during the delivery process. b. Propose two alternative delivery plans that account for the uncertainties identified above. c. List and explain the major causes of uncertainty. [Clearly state assumptions, if any]	3+3+4

Que. No.	Question	Max. Marks
Q4	Solve any TWO	20
ii)	Consider a problem instance: A home security system is designed to detect burglaries. The system includes an alarm that can be triggered by either a burglary or an earthquake. Two neighbours, John and Mary, may call the homeowner if they hear the alarm. However, each may or may not call depending on whether they hear the alarm and their own reliability. John and Mary do not hear the alarm perfectly every time. John confuses doorbell with alarm and Mary listens to loud music and thereby misses the alarms sometimes. a) Draw the Bayesian Network for the above problem. b) Assign probabilities for each node. [Assume suitable data] c) What is the probability that the alarm is triggered in the absence of a burglary or earthquake, and both John and Mary make calls? d) What is the probability that the alarm is not triggered, there is absence of a	2+4+2 +2

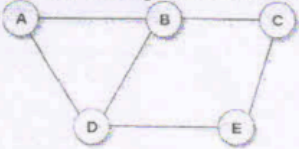
	burglary or but earthquake occurs, and both John and Mary miss to call.	
ii)	With neat sketch explains the architecture, characteristic features and roles of the expert system. Use some example to support your answers.	10
iii)	What is Action Description Language (ADL). Give ADL for drone delivery problem where two drones stationed at distinct locations exchange packages and transport them to their respective target locations. Follow ADL syntax; give clear initial state, goal state and Action descriptions.	2 + 8

Que. No.	Question	Max. Marks
Q5	Solve any FOUR	20
i)	Differentiate between planning and searching in Artificial Intelligence. Provide relevant examples.	5
ii)	Explain hierarchical task planning, use example: Constructing a villa after buying a plain land.	5
iii)	Discuss four approaches of AI in brief.	5
iv)	Write a short note on Natural Language Processing.	5
v)	Describe the architecture of a knowledge-based agent using the WUMPUS World example.	5
vi)	List and explain the forms of Learning.	5

Jan 24 - Apr 24

Que. No.	Question	Max. Marks
Q1	Solve any Four	20
i)	What are basic components of Intelligent systems?	5
ii)	Discuss and state the limitation of “ Thinking rationally approach of AI ”	5
iii)	Explain PEAS for automated driving scenario.	5
iv)	Discuss any two applications of AI.	5
v)	What is the advantage of Expert systems?	5
vi)	Give action description for wearing socks and shoes using ADL- Action Description Language.	5

Que No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	When to use Bayesian Networks? Draw topology and conditional probability table for grass being wet in the event of : a. Sprinkler being on, or b. It is raining or c. Sprinkler being on and it is raining, both. Assume suitable data	5
ii)	Compare and contrast between A* and AO* search algorithm, give a suitable example to support your answer.	5
	OR	
Q2 A	The goal here is to assign each letter a digit from 0 to 9 so that the arithmetic works out correctly. The rules are that all occurrences of a letter must be assigned the same digit, and no digit can be assigned to more than one letter. Use Constraint Satisfaction Problem with backtracking (CSP) to: A. Solve the given instance of crypt-arithmetic problem. B. State constraints clearly C. State heuristic chosen to solve the problem. D. Show step by step solution using the chosen heuristic.	10

	$ \begin{array}{r} \text{TWO} \\ + \text{TWO} \\ \hline \text{FOUR} \end{array} $	
Q 2 B	Solve any One	10
i)	Discuss alpha-beta pruning as improvement over min-max algorithm. Discuss how the values of alpha and beta are chosen for a two player game with a suitable example.	10
ii)	Consider an instance of graph coloring problem, wherein any two adjacent vertices cannot have the same color. Assuming the number of given colors =3, A. Solve the same using genetic algorithms. A sample chromosome could be: (1,2,1,2,1) wherein 1 and 2 are color numbers. You can use random numbers to assign the colors and use number of attacking pairs as fitness function. B. Show contents of all steps clearly. C. Compute answer till one generation.  Undirected Graph	10

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	What do you mean by Intelligent agents? Explain simple reflex agent as a solution for <u>predictive text suggestion in mobile keyboard</u> .	10
ii)	- Gita likes all kinds of food. - Mango and chapati are food. - Gita eats almond and is still alive. - Anything anyone eats and isn't killed by it is food. • Goal: Gita likes almond. Represent in First order logic and use forward chaining to prove the goal. State all inference rules used clearly.	10
iii)	Explain typical steps for Natural Language Processing.	10

Que. No.	Question	Max. Marks
Q4	Solve any Two	20
i)	Describe Hill Climbing Search algorithm. What are the problems faced by Hill Climbing Search ? Explain the solutions too.	10
ii)	Describe steps in knowledge engineering process with respect to detection of Covid-19 infection when symptoms are given.	10

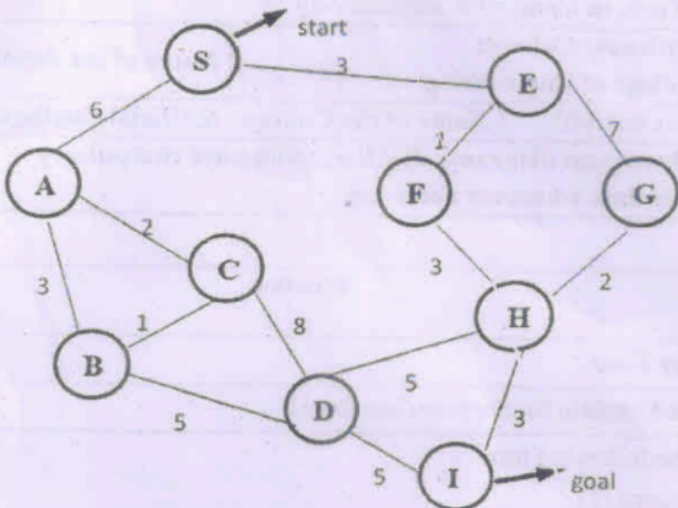
iii)	Design expert system for automated driving in mountains. Draw expert system component diagram and clearly state sample contents of them. State atleast 3 inference rules those might be part of the designed ES.	10
------	--	----

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any four	20
i)	Explain the properties of task environment in Artificial Intelligence.	5
ii)	Consider a problem of arranging rainy season trip to Lonavla for a joint family of 10, ranging from ages 1 to 70. The trip would be planned for mid-June. A. Give atleast 2 plans of trip. B. What could be the uncertainties in the above plan.	5
iii)	Differentiate between planning and searching.	5
iv)	Differentiate between Uninformed search and Informed search strategy in AI.	5
v)	Explain the terms universal quantifier and existential quantifier.	5
vi)	State and discuss the things those AI cannot do in technical forum.	5

Jan 23 - May 23

Que. No.	Question	Max. Marks
Q1	<i>Solve Any Four</i>	20
i)	Define and explain four approaches for AI.	5
ii)	Define the following term :- a) Agent (1) b) Rational agent (1) c) Omniscient Agent (1) d) Turing Test (2)	5
iii)	Write well-formed formulae for the following : a) Some students took French in spring 2001. (1.5) b) Every student who takes French passes it. (1.5) c) Only one student took Greek in spring 2001. (2)	5
iv)	Describe constraint satisfaction problems. While solving CSPs, which variable should be assigned next, and in what order should its values be tried?	5
v)	Differentiate between ADL and STRIPS language of planning.	5
vi)	Discuss the characteristics of expert system.	5

Que. No.	Question	Max. Marks
Q2 A	<i>Solve the following</i>	10
i)	Consider the Water Jug Problem “You are given two jugs, a 4-gallon one and 3-gallon one. Neither has any measuring marker on it. A pump which has unlimited water which you can use to fill the jug, and ground on which water may be poured. How can you get exactly 2 gallons of water in the 4-gallon jug?” Describe the operator and production rules.	5
ii)	Discuss the properties of task environment for part picking robot.	5
OR		

Q2 A	Sketch the architecture of utility agent. Discuss the component with the help of suitable example.	10
Q2 B	<i>Solve Any One</i>	10
Consider the graph given in Figure 1 below. Assume that initial state is S and goal state is I.  Figure 1		
i)	Draw the state space tree (4). Find a path from the initial state to the goal state with cost using the following algorithm. - Breadth First Search (BFS - 3) - Depth First Search (DFS - 3)	10
ii)	Assuming the heuristic function : $h(S)=10$, $h(A) = 8$, $h(B)=5$, $h(C)=7$, $h(D)=3$, $h(E)=6$, $h(F)=5$, $h(G)=3$, $h(H)=1$, $h(I)=0$. Use the A* Search to find a path from the initial state to the goal state with cost.	10

Que. No.	Question	Max. Marks
Q3	<i>Solve the following.</i>	20
i)	Explain the knowledge engineering process for one bit full adder circuit.	10
ii)	Consider the following set of sentences. - Whoever can read is literate. - Dolphins are not literate. - Some dolphins are intelligent. Prove/Goal - Some who is intelligent, cannot read. Convert the sentences in First Order Logic (4). Prove the sentence using resolution.(3+3)	10
OR		
ii)	Consider a new burglar alarm installed at home. It is fairly reliable at detecting a burglary, but also responds on occasion to minor earthquakes. There are two neighbours, John and Mary, who have promised to call the home owner at work	10

	when they hear the alarm. John always calls when he hears the alarm, but sometimes confuses the telephone ringing with the alarm and calls then, too. Mary, on the other hand, likes rather loud music and sometimes misses the alarm altogether. (Assume necessary data) - Draw topology of the network. - Associate a conditional probability table for each node.(2) - Given the evidence that neither a burglary nor an earthquake has occurred and both John and Mary call, Estimate the probability of ringing the alarm bell. (3) - What are the reasons that uncertainty gets introduced in problem solving?(3)	
--	---	--

Que. No.	Question	Max. Marks
Q4	Solve the following	20
i)	Explain the partial order planning algorithm for wearing shoes.	10

OR

i)	Discuss the Steps to build an NLP? Give applications of NLP in various areas.	10
ii)	Discuss the alpha-beta pruning algorithm. Comment on its strength and weaknesses. Solve the given problem instance using alpha-beta pruning to compute final score and declare the winner.	4+2+4

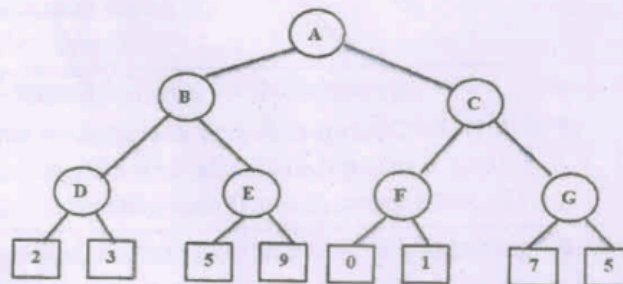


Figure 2

Que. No.	Question	Max. Marks
Q5	Solve Any Four.	20
i)	Define the terms chromosome (1), fitness function (1), crossover (2) and mutation (1) used in Genetic Algorithm.	5
ii)	Discuss the problems in hill climbing algorithm solutions.	5
iii)	Discuss the heuristic property admissibility with example.	5
iv)	Discuss the Universal Instantiation and Existential Instantiation Elimination with example.	5
v)	Describe the architecture of the Expert System with a diagram.	5
vi)	Discuss the Learning process in Decision Tree.	5

Question No.	SECTION A	Marks
Q1(a)	State and explain the "AI Problems". Brief on Mundane, Formal and Expert tasks. OR (i) List 5 things AI cannot do yet. (ii) List 5 applications of AI in daily life.	10M 5M 5M
Q1(b)	Describe each of the components of the Utility based agent. Support with a neat diagram. Comment on the example of a Mars lander on the surface of mars with obstacles in its way and how would it be different in a Goal based agent with learning.	3M 3M 4M
Q2(a)	What are Production rules? <u>OR Q.2 A Below</u> List the production rules for Water Jug Problem	2M 8M
Q2(b)	Define straight line distance as a heuristic. Distinguish between AO* Search Algorithm and A* search algorithm. Differentiate between BFS and DFS.	02M 04M 04M
Q3.1	Artificial Intelligence is about _____. 1. Playing a game on Computer 2. Making a machine Intelligent 3. Programming on Machine with your Own Intelligence 4. Putting your intelligence in Machine	1M
Q3.2	Who is known as the -Father of AI"? 1. Fisher Ada 2. Alan Turing 3. John McCarthy 4. Allen Newell	1M

Q. 2 A. Explain steps in genetic algorithms with a suitable example. 10M.
(2M for each step.)

Q3.3	Among the given options, which search algorithm requires less memory? 1. Optimal Search 2. Depth First Search 3. Breadth-First Search 4. Linear Search	1M
Q3.4	A technique that was developed to determine whether a machine could or could not demonstrate the artificial intelligence known as the ____ 1. Boolean Algebra 2. Turing Test 3. Logarithm 4. Algorithm	1M
Q3.5	Which algorithm is used in the Game tree to make decisions of Win/Lose? 1. Heuristic Search Algorithm 2. DFS/BFS algorithm 3. Greedy Search Algorithm 4. Min/Max algorithm	1M
Q3.6	An AI agent perceives and acts upon the environment using ____. 1. Sensors 2. Perceiver 3. Actuators 4. Sensors and Perceivers	1M
Q3.7	Which agent deals with the happy and unhappy state? 1. Utility-based agent 2. Model-based agent 3. Goal-based Agent 4. Learning Agent	1M
Q3.8	Which of the following is not an application of artificial intelligence? 1. Computer Vision 2. Natural Language Processing 3. Database Management System 4. Digital Assistants	1M
Q3.9	What is the work of Task Environment and Rational Agents? 1. Problem and Solution 2. Solution and Problem 3. Observation and Problem 4. Observation and Solution	1M

Q3.10	Machines that try to imitate human intuition while handling vague information lie in the field of AI called? 1.Functional Logic 2.Fuzzy Logic 3.Boolean Logic 4.Human Logic	IM
-------	---	----

	SECTION B	
Q 4(a)	Solve using CSP TWO+TWO=FOUR	10M
	OR	
Q 4(a)	Explain the steps of knowledge engineering process for a detective agent. Give sample output for every step.	10M
Q 4(b)	Translate the following set of sentences into FOL. Use backward chaining on the goal- Joel is at museum. a. Joel is a bulldog. b. John is Joel's master. c. The day is Saturday d. It is cold on Saturday. e. Joel is trained. f. Trained bulldogs are good dogs. g. If a dog is a good dog and has a master and at some place, then he will be with his master. h. If the day is Saturday and the day is cold, John is at the museum. <i>Give</i>	10M
Q 5(a)	STRIPS solution for the problem of changing flat tire with a spare one in car trunk. Clearly specify initial state, goal(s), actions and their pre-post conditions	10M
Q 5(b)	Differentiate between Supervised learning, Unsupervised learning, Semi supervised learning and reinforcement learning. OR List and explain the steps to build an NLP	10M 10M

Q 6.1	Which algorithm is used in the Game tree to make decisions of Win/Lose? 1. Heuristic Search Algorithm 2. DFS/BFS algorithm 3. Greedy Search Algorithm 4. Min/Max algorithm	1M
Q6.2	In the Wumpus World Problem, the reason for the uncertainty is that the agent's sensor gives only__ 1. Full & Global information 2. Partial & Global Information 3. Full & local information 4. Partial & local Information	1M
Q 6.3	The maximum depth to which the alpha-beta pruning can be applied. 1. Eight states 2. Six states 3. Ten states 4. Any depth	1M
Q 6.4	Face Recognition system is based on which type of approach? a) Weak AI approach b) Applied AI approach c) Cognitive AI approach d) Strong AI approach	1M
Q6..5	What does the Bayesian network provide? a) Partial description of the domain b) Complete description of the problem c) Complete description of the domain d) None of the mentioned	1M
Q6..6	Which of the following is an application of Artificial Intelligence? a) It helps to exploit vulnerabilities to secure the firm b) Language understanding and problem-solving (Text analytics and NLP) c) Easy to create a website d) It helps to deploy applications on the cloud	1M

Q6.7	The process by which the brain incrementally orders actions needed to complete a specific task is referred as _____ a) Planning problem b) Partial order planning c) Total order planning d) Both Planning problem & Partial order planning	1M
Q6.8	Following is/are the components of the partial order planning. a) Bindings b) Goal c) Causal Links d) All of the mentioned	1M
Q 6.9	Though local search algorithms are not systematic, key advantages would include _____ a) Less memory b) More time c) Finds a solution in large infinite space d) Less memory & Finds a solution in large infinite space	1M
Q6.10	What are the main cons of hill-climbing search? a) Terminates at local optimum & Does not find optimum solution b) Terminates at global optimum & Does not find optimum solution c) Does not find optimum solution & Fail to find a solution d) Fail to find a solution	1M

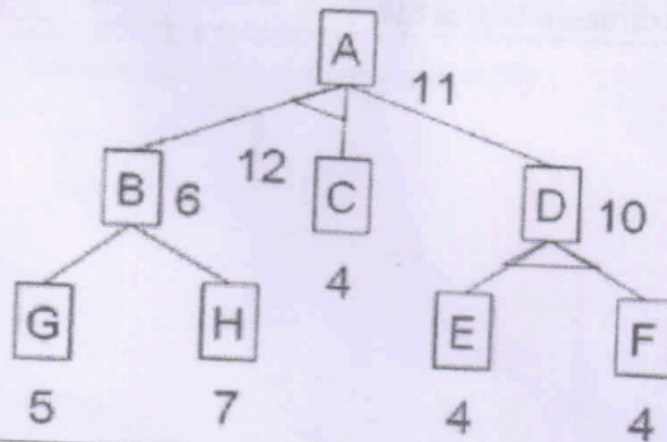
Q. 61. Human brain uses: for reasoning: —

- Forward chaining
- backward chaining
- Resolution
- Predicate logic

1 M.

May 22 - Jun 22

Question No.		Marks
Q 1 (a)	State and explain the "AI Problems". Brief on Mundane, Formal and Expert tasks. OR (i) List 5 things AI cannot do yet. (ii) List 5 applications of AI in daily life.	10M 5M 5M
Q 1 (b)	Describe each of the components of the Utility based agent. Support with a neat diagram. Comment on the example of a Mars lander on the surface of mars with obstacles in its way and how would it be different in a Goal based agent with learning.	3M 3M 4M
Q2 (a)	What are Production rules? Enumerate the steps to problem formulation in Block's world problem.	02M 08M
Q2 (b)	Define straight line distance as a heuristic. Distinguish between AO* Search Algorithm and A* search algorithm. Solve using AO* search algorithm	02M 03M 05M



Q3 (a)	Solve using CSP and show all steps clearly. TWO + TWO = FOUR OR 3A in Addendum	10M
Q3 (b)	Explain the steps of knowledge engineering process for a detective agent. Give sample output for every step.	10M
Q4 (a)	Brief on the following (i) Universal Generalization (ii) Universal Instantiation. (iii) Existential Instantiation (iv) Existential Introduction. OR 4A in Addendum	10M
Q4 (b)	Translate the following set of sentences into FOL. Use backward chaining on the goal- <u>Joel is at museum</u> . a. Joel is a bulldog. b. John is Joel's master. c. The day is Saturday d. It is cold on Saturday. e. Joel is trained. f. Trained bulldogs are good dogs. g. If a dog is a good dog and has a master and at some place, then he will be with his master. h. If the day is Saturday and the day is cold, John is at the museum. Give	10M
Q5 (a)	STRIPS solution for the problem of changing flat tire with a spare one in car trunk. Clearly specify initial state, goal(s), actions and their pre-post conditions	10M
Q5 (b)	Differentiate between Supervised learning, Unsupervised learning, Semi supervised learning and reinforcement learning. OR List and explain the steps to build an NLP	10M

Addendum to the question paper.

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
MAY-JUNE 2022

Class: TY B.Tech
Name of the Course: Artificial Intelligence
Course Code: 2UCC603

Semester: VI
Branch: Computer Engg.

Q. 3 A. Consider an instance of 0/1 knapsack as:

$N=6$, Capacity = 20, weight = $\{2,4,5,3,4,2\}$ and profit = $\{10,15,15,20,200,50\}$

Solve the problem using genetic algorithms.

- Clearly state thresholds or any other assumptions
- Show step by step output after every step.

Compute solution upto only ONE generation.

Q. 4 A Consider the Lung Cancer Problem- Pollution and smoking are the factors those affect a patient's chance of having cancer. Similarly, having cancer affects the patient's breathing and the chances of having a positive X-ray result.

- Draw Bayesian network for the above stated problem 2M
- Assign probabilities to every node in the network 2M
- Compute probability of a smoker having lung cancer, has a positive X-ray result but faces no problem in breathing. 3M
- State reasons of uncertainty in general. 3M

Question No.		Max Marks
Q1 (A)	Solve the following multiple choice questions. - Which one of the following statements is true? - BFS runs out of memory before it runs out of time. - BFS runs out of time before it runs out of memory. - Typically, BFS needs less memory than DFS for the execution - BFS is good choice when the branching factor is large - For a given problem, there could be ____ partial order plan(s), while there could be ____ total order plan(s). - One, many - Many, many - Many, one - One, one - Choose the correct FOL representation for the given statement. <u>No mortal can live for more than 150 years.</u> $\forall m, t1, t2: \text{mortal}(m) \wedge \text{born}(m, t1) \wedge \text{greater_than}(t2 - t1, 150) \rightarrow \text{dead}(m, t2)$ $\forall m, t1, t2: \text{mortal}(m) \wedge \text{born}(m, t1) \wedge \text{greater_than}(t2 - t1, 150) \rightarrow \sim \text{alive}(m, t2)$ $\sim \exists m, t1, t2: \text{mortal}(m) \wedge \text{born}(m, t1) \wedge \text{greater_than}(t2 - t1, 150) \wedge \text{dead}(m, t2)$ $\sim \exists m, t1, t2: \text{mortal}(m) \wedge \text{born}(m, t1) \wedge \text{greater_than}(t2 - t1, 150) \wedge \sim \text{alive}(m, t2)$ - Consider two chromosomes, X1: ABCDEFG and Y1: ZYXVUTS. If one chooses two point crossover as 3:3:1 then the new chromosomes after the crossover would be: - X1' :AYCVDETG Y1': ZBXDUFS - X1' : ZYXDEFS Y1': ABCVUTG - X1' :ABCVUTG Y1': ZYXDEFS - X1' :AYXVEFG Y1': ZBCDUTS - A problem may have _____ heuristic(s). - Only one - Many - Maximum two - Maximum 5	10

	6. Which one of the following statements is true? - Every rule based reasoning system is also a case based reasoning system, but the reverse isn't always true. - Case based reasoning system and rule based reasoning system are totally different; they don't have subset-superset relationship between them. - Every case based reasoning system and rule based reasoning system are mutually exclusive. - Every case based reasoning system is also a rule based reasoning system, but the reverse isn't always true. 7. Properties of knowledge are: - Hard to characterize, voluminous, constantly changing. - Unpredictable, hard to measure, constantly changing. - Laziness of collecting knowledge, practical and conceptual ignorance - Hard to represent, hard to collect, hard to maintain 8. Human mind uses _____ for reasoning process. - Forward chaining - Backward chaining - Resolution - Proof by refutation 9. The snakes and ladders belong to _____ category of games. - Deterministic, static, strict alternate, multiplayer - Non-Deterministic, static, no-strict alternate, multiplayer - Non-Deterministic, dynamic, no-strict alternate, multiplayer - Deterministic, dynamic, no-strict alternate, multiplayer 10. The most appropriate agent architecture for a psychological counseling agent would be: - Learning with goal based architecture as performance element - Learning with utility based architecture as performance element - Learning with reflex based architecture as performance element - Learning agent architecture	
Q1 (B)	Attempt any FIVE questions out of the following (any 5 out of 7) - Explain the concept of alpha and beta values in adversarial search with a suitable example. - Discuss the "Thinking rationally approach". State its limitations. - Discuss the sequence in which variable-constraint assignments are considered by CSP algorithms to improve the backtracking efficiency? - List various algorithm families those could be used by an expert system's inference engine. - State which one of the problem characteristics are applicable to Tower of Hanoi problem. - State and discuss the things those AI cannot do even in today's technically developed era. - Discuss applications of natural language processing.	10

Q. 2	Consider following set of statements. 1. Lucy is a professor 2. All professors are people. 3. John is the dean. 3. Deans are professors. 4. All professors consider the dean a friend or don't know him. 5. Everyone is a friend of someone. 6. People only criticize people that are not their friends. 7. Lucy criticized John. Represent the statements in FOL. Prove: Lucy and John aren't friends using forward chaining OR backward chaining.	10
Q. 3	A. Consider a plan of getting COVID 19 vaccine at the vaccination center at in the nearby hospital. Right from scheduling appointment to actually getting the vaccine, a. List down uncertainties in the plan. b. Give at least two plans considering the above mentioned uncertainties. B. Consider a Life expectancy Problem-It is known that whether or not a person has cancer, is directly influenced by whether (s)he is exposed to second-hand smoke and whether (s)he smokes. Both of these things are affected by whether her parents smoke. Cancer reduces a person's life expectancy. a. Draw Bayesian network for the abovestated problem b. Assign probabilities to each node in the network. c. Compute probability of reduction in life expectancy if the person's parents are nonsmoker, but the person is a smoker and is suffering from lung cancer.	4 + 6
Q. 4	A. Give ADL description for wearing socks and shoes problem. B. Differentiate between STRIPS and ADL. C. Comment on strengths and weaknesses of decision trees.	4+3+3